

Slide 1 - Machine Learning Foundations

Hello everyone, and welcome to BLA 1, Part 2 of my AWS Machine Learning Engineer - Associate series. Part 1 introduced AWS cloud foundations. In this presentation, I will introduce the machine learning concepts that we need before mapping the complete lifecycle to AWS services. The objective is to understand the reasoning behind machine learning rather than to build or train a model.

I will begin with common machine learning problem types, including classification, regression, clustering, and generation. I will then compare traditional machine learning with foundation models. After that, I will explain the learning lifecycle, the difference between training and inference, the importance of choosing appropriate evaluation metrics, and four major model-quality risks: underfitting, overfitting, bias, and drift.

These concepts appear throughout the MLA-C02 exam. Certification questions frequently present a business requirement and ask which modeling approach, metric, deployment method, or monitoring strategy is most appropriate. Memorizing service names is not enough. We need to understand what type of problem is being solved, what output is expected, what error is most costly, and how the system may change after deployment. By the end of this presentation, we should be able to describe machine learning as a managed and repeatable cycle rather than as a single algorithm.

Slide 2 - The business question determines the learning approach

The first step in machine learning is identifying the type of business question. Classification predicts a category. Examples include determining whether a transaction is fraudulent, whether an image contains a defect, or whether a customer is likely to cancel a subscription.

Regression predicts a numeric value. Examples include estimating the selling price of a house, forecasting delivery time, or predicting energy consumption. The important distinction is that the result is a continuous number rather than a named category.

Clustering is an unsupervised learning task. It identifies groups in data when predefined labels are not available.

Generation creates new content such as text, images, audio, summaries, or code. Generative AI commonly uses foundation models that respond to prompts and optional context.

Slide 3 - Traditional ML specializes; foundation models generalize

Traditional machine learning models are usually developed for a specific prediction task. A team collects relevant data, selects features, chooses an algorithm, trains the model, and evaluates its performance.

Retrieval Augmented Generation, often called RAG, connects a foundation model with an external knowledge source. The system retrieves relevant information and places it into the model's context before generating an answer.

Slide 4 - A model moves through a repeatable learning cycle

Machine learning follows a lifecycle that begins with defining the problem. A vague goal such as improving customer experience is not yet a machine learning objective. The team must identify the prediction or output, the intended user, the decision the output will support, and a measurable definition of success.

The next stage is data preparation. Data may need to be collected from multiple sources, cleaned, labeled, transformed, and divided into training, validation, and test sets.

During the learning stage, a traditional algorithm is trained or a foundation model is adapted. Engineers select hyperparameters, compute resources, stopping criteria, and optimization methods.

After evaluation, the model is deployed for inference. Deployment does not end the lifecycle. Real-world data changes, users behave differently, costs fluctuate, and systems fail. Monitoring can reveal drift, poor output, unusual latency, or resource problems. The lifecycle is therefore circular. Each stage produces information that can improve an earlier stage, and good governance keeps those changes controlled and repeatable.

Slide 5 - Training creates behavior; inference uses it

Training and inference are two different types of machine learning workload. Training is the process that creates or adapts model behavior. In traditional machine learning, an algorithm examines examples and adjusts internal parameters so that it can identify patterns

Inference occurs when a trained or selected model receives new input and returns an output. A classifier returns a class or probability. A regression model returns a number. A foundation model may return generated text or another media type. Inference architecture depends on how quickly the result is needed and how much traffic is expected.

Slide 6 - Evaluation metrics must reflect the cost of error

Model evaluation must match both the type of model and the consequence of error. For classification, accuracy measures the proportion of predictions that are correct, but accuracy can be misleading when classes are imbalanced.

Precision measures how many predicted positive cases are actually positive. High precision is important when false positives are expensive.

Regression metrics measure numeric error. Mean absolute error is easy to interpret because it averages the absolute difference between predicted and actual values.

Generative AI evaluation is broader. A response may need to be relevant, accurate, grounded in supplied information, safe, unbiased, and stylistically appropriate.

Production fitness adds operational metrics such as latency, throughput, availability, drift, and cost. The key exam strategy is to identify what kind of mistake matters most. The correct metric is the one that represents the real consequence described in the scenario, not simply the metric that is most familiar.

Slide 7 - Model quality can fail before or after deployment

Model quality can fail in several different ways, and each failure requires a different response.

Underfitting occurs when a model is too simple, insufficiently trained, or unable to capture useful patterns.

Overfitting occurs when a model learns training details too closely and fails to generalize to unfamiliar data. It may perform extremely well on training examples but poorly on validation or test data.

Bias refers to systematic problems in data or model behavior that create unfair or unrepresentative outcomes.

Drift occurs after deployment when real-world conditions change. Data drift means the input distribution changes.

The major takeaway from Part 2 is that machine learning is a managed lifecycle, not a single algorithm. We begin with a measurable objective, select an approach that fits the data and output, evaluate the consequences of error, deploy appropriately, and monitor change. Part 3 will map this lifecycle to the four official MLA-C02 domains and show how the complete seven-BLA series will progress.